A positive-dimensional fixed character locus in genus three

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Abstract

We construct an algebraic family of rank-81 characters of a closed genus-three surface, all fixed by the same finite-index subgroup of its mapping class group. Every member is semisimple, and all but finitely many members are irreducible. The family has trivial determinant and an explicit nonconstant trace. In particular, it contradicts both the finiteness and the integrality assertions in Litt's Conjecture 6.1.7. The construction also gives a counterexample to the Putman–Wieland conjecture for closed genus-three surfaces. We start with an explicit cocycle for a finite Heisenberg representation. A finite-normalizer argument produces an invariant integral cohomology class on a degree-81 cover; exponentiating and inducing this class gives the character family. We also construct the resulting algebraic isomonodromic sections over a common finite cover of moduli space.

1 Introduction

Let Σ_g be a closed, connected, oriented surface of genus g, and put $\Pi_g = \pi_1(\Sigma_g)$. Its orientation-preserving mapping class group Mod_g acts on the character variety

$$X_r(\Sigma_g) = \mathrm{Hom}(\Pi_g, \mathrm{GL}_r(\mathbf{C})) /\!\!/ \mathrm{GL}_r(\mathbf{C}).$$

The complex points of this variety classify semisimple representations up to isomorphism. We write $X_r(\Sigma_g)^{\mathrm{irr}}$ for the irreducible locus. Since inner automorphisms of Π_g act trivially on characters, the mapping class group action is independent of the choice of a base point.

A character has finite mapping class group orbit precisely when it is fixed by some finite-index subgroup of Mod_g. We construct a nonconstant family fixed by the same finite-index subgroup.

Theorem 1.1. *There exist a finite-index subgroup $\Gamma < \mathrm{Mod}_3$ and an algebraic family*

$$R_t : \Pi_3 \longrightarrow \mathrm{SL}_{81}(\mathbf{C}), \qquad t \in \mathbf{C}^\times,$$

with the following properties.

1. *Every R_t is semisimple. There is a finite set E of roots of unity such that R_t is irreducible for $t \notin E$.*
2. *The same subgroup Γ fixes $[R_t]$ for every t.*
3. *For the standard surface generators a_i, b_i,*

$$\mathrm{tr}\, R_t([a_1, a_2]) = 9(t + t^{-1} + 1)^2. \tag{1.1}$$

In particular, $X_{81}(\Sigma_3)^{\mathrm{irr},\Gamma}$ contains a positive-dimensional algebraic family.

The matrices of R_t are monomial: each row and column has exactly one nonzero entry, and

every nonzero entry is an integral power of t. Thus the family is defined over $\mathbf{Z}[t, t^{-1}]$.

Litt's Conjecture 6.1.7 predicts that, for $g \geq 3$, the irreducible characters fixed by any given finite-index subgroup form a finite set, and that every such character is defined over the ring of integers of a number field [3, Conjecture 6.1.7]. Theorem 1.1 contradicts both predictions.

Corollary 1.2. *Both assertions of [3, Conjecture 6.1.7] fail for $(g, n, r) = (3, 0, 81)$. In particular, there is an irreducible $\mathrm{SL}_{81}(\mathbf{Q})$ representation with finite Mod_3-orbit which cannot be defined over the ring of integers of any number field.*

Proof. The nonconstant trace in (1.1) proves the failure of finiteness on the irreducible locus. For integrality, take $t = 2$. This parameter is not a root of unity, so R_2 is irreducible, and

$$\operatorname{tr} R_2([a_1, a_2]) = 9\left(2 + \frac{1}{2} + 1\right)^2 = \frac{441}{4}.$$

A representation defined over a number ring has algebraic-integer traces. The rational number $441/4$ is not an algebraic integer. □

The construction has three steps. First, we give a nonzero cocycle z for a 27-dimensional representation of Π_3 with finite Heisenberg image. Explicit identities for a full set of mapping class group generators show that the cohomology class of z is preserved after identifying the coefficient representations. These identifications lie in a finite group. Passing to one finite-index subgroup therefore makes both the coefficient representation and $[z]$ fixed.

Second, a linear functional on the coefficient space converts z into a nonzero integral class β on a degree-81 cover of Σ_3. The same subgroup fixes β. Finally, we exponentiate this class, obtaining the rank-one character t^{β}, and induce it to Π_3. The general mechanism is isolated in Section 2; the remaining work is an explicit construction of its input.

The resulting cover also has a nonzero cohomology class with finite mapping class group orbit; see Corollary 5.2. The algebraic isomonodromic interpretation is given in Section 6. The main group-theoretic proof is independent of these geometric consequences.

The theorem concerns genus three. It does not establish an analogous construction for every $g \geq 4$, nor does it settle the question for sufficiently large genus posed in [2, Question 10.1.3]. Adding trivial summands gives semisimple examples in every rank at least 81, but our irreducible family has rank 81.

2 From invariant cohomology to fixed characters

An integral cohomology class on a connected cover is a homomorphism from its fundamental group to $\mathbf{Z}$. Exponentiation turns it into a family of rank-one representations. The following elementary observation explains why induction is useful here.

Proposition 2.1. *Let Π be a finitely generated group, let $N \lhd \Pi$ have finite index, and let $N \subset H \subset \Pi$. Let $\beta : H \to \mathbf{Z}$ be a homomorphism. Suppose that a subgroup $\mathcal{A} \subset \mathrm{Aut}(\Pi)$ preserves both H and β. Set*

$$\chi_t(h) = t^{\beta(h)}, \qquad R_t = \mathrm{Ind}_H^{\Pi} \chi_t, \qquad d = [\Pi : H].$$

Then R_t is an algebraic family of semisimple monomial representations of rank d, and $\mathcal{A}$ fixes $[R_t]$ for every t. If $\beta \neq 0$, the family of characters is nonconstant. If, in addition, the stabilizer of $\beta|_N$ under conjugation by Π is exactly H, then R_t is irreducible outside a finite set of roots of unity.

Proof. Choose representatives x of the left cosets Π/H. In the corresponding induced basis, each matrix entry is either zero or a power of t. Since N is normal and contained in H, restriction to N is diagonal, with characters

$$n \longmapsto t^{\lambda_x(n)}, \qquad \lambda_x(n) = \beta(x^{-1}nx). \tag{2.1}$$

The homomorphism λ_x depends only on the coset xH.

This restriction is semisimple for every t. Given a Π-invariant subspace, choose an N-equivariant projection onto it and average the projection over Π/N. Conjugation by N fixes the projection, so the average is well defined and Π-equivariant. Every invariant subspace therefore has an invariant complement.

Under the stabilizer hypothesis, the homomorphisms λ_x are pairwise distinct. For each pair of distinct cosets, choose an element of N on which their exponents differ. Their characters can coincide only when $t^m = 1$ for a corresponding nonzero integer m. The union of these exceptional sets is finite. At any other parameter, the N-weight spaces are distinct lines. The group Π permutes these lines transitively, so the representation is irreducible.

An automorphism in $\mathcal{A}$ preserves the inducing pair (H, χ_t), and consequently preserves the induced character. Finally, if $\beta \neq 0$, then $\beta|_N \neq 0$, since N has finite index in H. Choose $n \in N$ with $\beta(n) \neq 0$. The trace

$$\operatorname{tr} R_t(n) = \sum_{xH \in \Pi/H} t^{\lambda_x(n)}$$

is a nonconstant Laurent polynomial: all coefficients are nonnegative, and at least one exponent is nonzero. □

We will also use a direct criterion for the determinant to be trivial. In the notation of the proof, write

$$\det R_t(\gamma) = \varepsilon(\gamma) t^{d_0(\gamma)}. \tag{2.2}$$

Here ε is the sign of the coset permutation, and $d_0 : \Pi \to \mathbf{Z}$ is a homomorphism. On N it is given by

$$d_0(n) = \sum_{xH \in \Pi/H} \lambda_x(n).$$

If this sum is zero, then $d_0 = 0$, because N has finite index in Π. If the permutation image has odd order, then also $\varepsilon = 1$. These two conditions imply that the entire family takes values in SL_d.

3 A Heisenberg cocycle in genus three

We now construct the invariant class needed in Proposition 2.1. Throughout the rest of the paper,

$$\Pi = \Pi_3 = \langle a_1, b_1, a_2, b_2, a_3, b_3 \mid [a_1, b_1][a_2, b_2][a_3, b_3] = 1 \rangle ,$$

where $[a, b] = aba^{-1}b^{-1}$. Fix a primitive cube root of unity ζ, and put $F = \mathbf{Q}(\zeta)$ and $\mathfrak{o} = \mathbf{Z}[\zeta]$.

3.1 The finite coefficient representation

Let $U = F^3$ with basis e_0, e_1, e_2, indexed modulo three. Define

$$Ae_x = e_{x+1}, \qquad Be_x = \zeta^{-x} e_x.$$

Then $A^3 = B^3 = I$ and $[A, B] = \zeta I$. On $V = U^{\otimes 3}$, write A_i, B_i for these operators on the ith factor. Operators on distinct factors commute. Consequently

$$\rho(a_i) = A_i, \qquad \rho(b_i) = B_i \tag{3.1}$$

defines a representation of Π, since the surface relator maps to $\zeta^3 I = I$.

Its image is the finite Heisenberg group

$$G = \{\zeta^c A_1^{r_1} A_2^{r_2} A_3^{r_3} B_1^{s_1} B_2^{s_2} B_3^{s_3} : c, r_i, s_i \in \mathbf{F}_3\}. \tag{3.2}$$

These expressions are distinct, so $|G| = 3^7$. The representation is absolutely irreducible: the simultaneous B_i-eigenspaces are the tensor-basis lines, and the A_i permute those lines transitively.

3.2 The cocycle

Our convention for a cocycle is

$$z(gh) = z(g) + \rho(g) z(h), \qquad \delta w(g) = (\rho(g) - I) w.$$

Put

$$u = e_0 - \zeta e_1, \qquad v = u^{\otimes 3}, \qquad c_1 = v, \quad c_2 = -\zeta^2 v, \quad c_3 = 0, \tag{3.3}$$

and prescribe

$$z(a_i) = (A_i - I) c_i, \qquad z(b_i) = (B_i - I) c_i. \tag{3.4}$$

Lemma 3.1. *These values define an $\mathfrak{o}^{27}$-valued cocycle on Π. Its cohomology class is nonzero.*

Proof. On the subgroup generated by a_i, b_i, the cocycle is the coboundary δc_i. Hence

$$z([a_i, b_i]) = (\zeta - 1) c_i.$$

The value on the surface relator is

$$(\zeta - 1)(c_1 + \zeta c_2 + \zeta^2 c_3) = (\zeta - 1)(v - \zeta^3 v) = 0.$$

It therefore descends from the free group to Π. If $z = \delta w$, its vanishing on a_3, b_3 would force $A_3 w = B_3 w = w$. Their commutator is ζI, so $w = 0$. This is impossible because $z(a_1) \neq 0$. $\square$

The choice of the three c_i can be understood without guessing their coefficients. For any triple $(\lambda_1, \lambda_2, \lambda_3)$ satisfying

$$\lambda_1 + \zeta \lambda_2 + \zeta^2 \lambda_3 = 0, \tag{3.5}$$

the same construction with $c_i = \lambda_i v$ gives a cocycle. Adding the same scalar to all three λ_i changes it by a coboundary. Thus these cocycles give a one-dimensional space of cohomology classes. The choice $(1, -\zeta^2, 0)$ is convenient because one handle vanishes and the resulting integral period is primitive.

4 Mapping class symmetries

This section proves the essential global assertion: one finite-index subgroup fixes both ρ and $[z]$. We first give the generator identities, then explain why they imply finite-index invariance.

4.1 Generators and intertwiners

For $i = 1, 2, 3$, define automorphisms α_i, ν_i by

$$\alpha_i : a_i \longmapsto a_i b_i, \qquad \nu_i : b_i \longmapsto b_i a_i,$$

fixing every other generator. For $k = 1, 2$, set

$$\Theta_k = a_{k+1} b_{k+1}^{-1} a_{k+1}^{-1} b_k$$

and define τ_k by

$$a_k \longmapsto a_k \Theta_k, \qquad b_k \longmapsto \Theta_k^{-1} b_k \Theta_k, \qquad a_{k+1} \longmapsto \Theta_k^{-1} a_{k+1}, \tag{4.1}$$

fixing the remaining generators.

These are the Lickorish generators in the conventions of [1, Lemma 6.2.1 and Table 3]; the ν_i are left twists and the others are right twists. The eight twists generate the mapping class group of a genus-three surface with one boundary component. Capping that boundary by a disk with a marked point gives a surjection onto $\mathrm{Mod}_{3,1}$. We therefore regard the subgroup $\mathcal{A} < \mathrm{Aut}(\Pi)$ generated by the eight displayed automorphisms as $\mathrm{Mod}_{3,1}$, acting on the based surface group.

For $x = (x_1, x_2, x_3) \in \mathbf{F}_3^3$, define

$$\begin{aligned} D_i e_x &= \zeta^{-x_i(x_i-1)/2} e_x, \\ Q_i &= \frac{I + \zeta^2 A_i + A_i^2}{1 - \zeta}, \\ M_k e_x &= \zeta^{(x_k - x_{k+1})(x_k - x_{k+1}+1)} e_x. \end{aligned} \tag{4.2}$$

The exponents are read modulo three; $x_i(x_i - 1)/2$ may equally be evaluated on integer representatives.

Lemma 4.1. *The operators in* (4.2) *satisfy*

$$\rho \circ \alpha_i = D_i \rho D_i^{-1}, \qquad D_i^{-1}(z \circ \alpha_i) = z, \tag{4.3}$$

$$\rho \circ \nu_i = Q_i \rho Q_i^{-1}, \qquad Q_i^{-1}(z \circ \nu_i) = z, \tag{4.4}$$

$$\rho \circ \tau_k = M_k \rho M_k^{-1}. \tag{4.5}$$

For the cocycle under the two remaining twists,

$$\begin{aligned} &M_1^{-1}(z \circ \tau_1) = z, \\ &M_2^{-1}(z \circ \tau_2) = z + \delta w, \qquad w = (\zeta - \zeta^2) u \otimes e_1 \otimes e_0. \end{aligned} \tag{4.6}$$

All eight intertwiners belong to $\mathrm{SL}(V)$.

Proof. On one tensor factor, $Du = u$, $Qu = u$,

$$DAD^{-1} = AB, \qquad DBD^{-1} = B, \qquad QAQ^{-1} = A, \qquad QBQ^{-1} = BA.$$

The first identity for D follows by taking the ratio of consecutive diagonal entries. For Q, put $p(X) = 1 + \zeta^2 X + X^2$. Then

$$p(A)B = Bp(\zeta A), \qquad p(\zeta A) = Ap(A), \qquad p(A)u = (1-\zeta)u.$$

These identities give $QB = BAQ$ and $Qu = u$; since Q is a polynomial in A, it commutes with A. These give (4.3)–(4.4), since the cocycle on each handle is a coboundary with potential a scalar multiple of v.

The mixed twists require only a calculation on two tensor factors. Appendix A supplies that calculation in a uniform form. It proves (4.5) and, for the general cocycle with potentials $c_i = \lambda_i v$ satisfying (3.5), gives

$$M_k^{-1}(z \circ \tau_k) = z + \delta\big(\lambda_j(M_k^{-1}v - v)\big), \qquad \{j, k, k+1\} = \{1, 2, 3\}.$$

For $(\lambda_1, \lambda_2, \lambda_3) = (1, -\zeta^2, 0)$, this is exactly (4.6).

For completeness, the determinant checks are immediate. The nontrivial diagonal entry ζ^2 occurs nine times in each D_i and each M_k. On one factor, Q has eigenvalues $1, 1, \zeta$, so $\det Q_i = \zeta^9 = 1$. □

4.2 One subgroup fixes the actual class

The following lemma separates the finite-index argument from the explicit calculations. In particular, it does not require chosen intertwiners to satisfy mapping class group relations.

Lemma 4.2. *Let $\mathcal{A} < \mathrm{Aut}(\Pi)$ be generated by $\sigma_1, \dots, \sigma_m$, and let $\rho : \Pi \to \mathrm{GL}(V)$ and $z \in Z^1(\Pi, V)$ be given. Suppose a finite group $\mathcal{N} < \mathrm{GL}(V)$ contains operators P_i such that*

$$\rho \circ \sigma_i = P_i \rho P_i^{-1}, \qquad [P_i^{-1}(z \circ \sigma_i)] = [z].$$

Then a finite-index subgroup $\mathcal{A}_0 < \mathcal{A}$ fixes the literal homomorphism ρ and the class $[z]$.

Proof. Let $E < \mathcal{A} \times \mathcal{N}$ be generated by the pairs (σ_i, P_i). The two displayed properties are closed under products and inverses. For the cohomology assertion this follows from

$$T_{\sigma,P}(c) = P^{-1}(c \circ \sigma), \qquad T_{\sigma\circ\tau,PQ} = T_{\tau,Q}T_{\sigma,P}.$$

Thus every pair in E preserves $[z]$ in the indicated sense. The kernel of $E \to \mathcal{N}$ has finite index. Its image $\mathcal{A}_0$ in $\mathcal{A}$ also has finite index, since $E \to \mathcal{A}$ is surjective. For an element of this kernel, the coefficient operator is I, which gives the conclusion. □

Apply the lemma with

$$\mathcal{N} = N_{\mathrm{SL}(V)}(G).$$

This normalizer is finite. Indeed, conjugation maps it to the finite group $\mathrm{Aut}(G)$; by absolute irreducibility and Schur's lemma, the kernel consists of scalar matrices of determinant one, hence lies in μ_{27}. Lemma 4.1 shows that all eight intertwiners belong to this normalizer.

We have therefore obtained a finite-index subgroup $\widehat{\Gamma} < \mathrm{Mod}_{3,1}$ satisfying

$$\rho \circ \sigma = \rho, \qquad z \circ \sigma - z = \delta w_\sigma \quad (\sigma \in \widehat{\Gamma}) \tag{4.7}$$

for suitable $w_\sigma \in V$. Its image $\Gamma < \mathrm{Mod}_3$ has finite index as well.

5 The covering class and the character family

5.1 A primitive integral class

Let

$$K = \langle B_1, B_2, B_3 \rangle < G, \qquad H = \rho^{-1}(K), \qquad N = \ker \rho.$$

Since $|K| = 3^3$, the subgroup H has index 81 in Π. It corresponds to a connected unramified cover

$$Y \longrightarrow \Sigma_3$$

of degree 81, whose genus is $1 + 81(3-1) = 163$.

Let $\ell = e^*_{000} : V \to F$. The group K fixes ℓ. For $h \in H$, define

$$\widetilde{\beta}(h) = \ell z(h), \qquad \beta(h) = \mathrm{Tr}_{F/\mathbf{Q}}\big(\ell z(h)\big). \tag{5.1}$$

Because $\ell\rho(h) = \ell$ on H, the cocycle identity shows that $\widetilde{\beta} : H \to \mathfrak{o}$ and $\beta : H \to \mathbf{Z}$ are homomorphisms. Thus β is an integral cohomology class on Y.

A useful test element is

$$h_0 = [a_1, a_2] \in N.$$

The two coefficient operators A_1, A_2 commute. Using $c_2 - c_1 = \zeta v$, we find

$$z(h_0) = \zeta(A_1 - I)(A_2 - I)v, \qquad \ell z(h_0) = \zeta, \qquad \beta(h_0) = -1. \tag{5.2}$$

In particular, β is nonzero and primitive.

Every $\sigma \in \widehat{\Gamma}$ preserves H because it fixes ρ. Applying ℓ to (4.7) on H kills the coboundary term. Hence

$$\beta \circ \sigma = \beta \qquad (\sigma \in \widehat{\Gamma}). \tag{5.3}$$

This is the invariant integral class required in Proposition 2.1.

5.2 The stabilizer and the determinant

To prove irreducibility, it remains to show that the stabilizer of $\beta|_N$ is exactly H. We give a direct argument using the trace pairing of $F/\mathbf{Q}$.

For $n \in N$ and $\gamma \in \Pi$, the cocycle identity gives

$$z(\gamma^{-1} n \gamma) = \rho(\gamma)^{-1} z(n). \tag{5.4}$$

The F-linear span of $z(N)$ is therefore G-stable. It is nonzero by (5.2), and hence equals V by irreducibility.

Suppose γ stabilizes $\beta|_N$, and put $f = \ell\rho(\gamma)^{-1} - \ell$. Then

$$\mathrm{Tr}_{F/\mathbf{Q}}(f z(n)) = 0 \qquad (n \in N).$$

Choose $c \in \Pi$ with $\rho(c) = \zeta I$, for example $c = [a_1, b_1]$. Replacing n by cnc^{-1} gives $\mathrm{Tr}_{F/\mathbf{Q}}(\zeta f z(n)) = 0$ as well. The trace pairing is nondegenerate and $1, \zeta$ form a $\mathbf{Q}$-basis of F, so $f z(n) = 0$ for all n. Since $z(N)$ spans V, we conclude that $f = 0$.

The stabilizer of ℓ in G is exactly K: a nonzero A-shift moves its support, while a nontrivial central scalar changes its value. Thus $\gamma \in H$. Conversely, H stabilizes $\beta|_N$ because β is a homomorphism on H. This proves the required stabilizer assertion.

For the determinant, identify Π/H with G/K. The central subgroup $\langle \zeta I \rangle$ acts freely on this set in orbits of length three. Along each such orbit, the sum of the exponent homomorphisms in (2.1) is zero, since

$$\mathrm{Tr}_{F/\mathbf{Q}}(a) + \mathrm{Tr}_{F/\mathbf{Q}}(\zeta a) + \mathrm{Tr}_{F/\mathbf{Q}}(\zeta^2 a) = 0.$$

Hence the sum of all the λ_x vanishes on N. The permutation image has odd order, being a quotient of G. The criterion (2.2) now gives trivial determinant.

5.3 Proof of the main theorem and its trace formula

Define

$$R_t = \mathrm{Ind}_H^{\Pi}(t^\beta). \tag{5.5}$$

Proposition 2.1 and the preceding stabilizer calculation prove semisimplicity for all t, irreducibility away from finitely many roots of unity, and nonconstancy of the character family. The determinant calculation shows that R_t takes values in SL_{81}. Equation (5.3) shows that the same $\widehat{\Gamma}$ fixes every induced character. Inner automorphisms act trivially on characters, so its image $\Gamma < \mathrm{Mod}_3$ fixes the entire family.

It remains to compute the trace in (1.1). We have

$$(A - I)u = -e_0 - \zeta^2 e_1 - \zeta e_2.$$

Thus the coordinate of $z(h_0)$ at (x_1, x_2, x_3) is

$$\begin{cases} \zeta^{\,1-x_1-x_2}, & x_3 = 0, \\ -\zeta^{\,2-x_1-x_2}, & x_3 = 1, \\ 0, & x_3 = 2. \end{cases}$$

Representatives of G/K may be taken to be $\zeta^c A_1^{r_1} A_2^{r_2} A_3^{r_3}$. For each of the nine coordinates in the first row, multiplication by the three central scalars gives field traces $2, -1, -1$. The nine coordinates in the second row give $-2, 1, 1$. The nine zero coordinates contribute 27 zero exponents. Consequently the exponent multiplicities of $R_t(h_0)$ are

exponent	-2	-1	0	1	2
multiplicity	9	18	27	18	9.

It follows that

$$\mathrm{tr}\, R_t(h_0) = 9t^{-2} + 18t^{-1} + 27 + 18t + 9t^2 = 9(t + t^{-1} + 1)^2.$$

This completes the proof of Theorem 1.1.

Remark 5.1. For $|t| \neq 1$, the representation R_t has infinite image and is not unitarizable. Indeed, the displayed eigenvalues of $R_t(h_0)$ include t^2 and t^{-2}. Thus the family is not explained by varying representations of a finite group.

5.4 The finite-cover homology consequence

The construction also gives a direct finite-orbit statement in the homology of a finite cover. We record it on the regular cover to avoid any dependence on conventions about non-Galois covers.

Corollary 5.2. *Let $\widetilde{Y} \to \Sigma_3$ be the regular cover corresponding to $N = \ker \rho$. It has degree 3^7 and genus 4375. The action of the stabilizer of N in $\mathrm{Mod}_{3,1}$ on $H^1(\widetilde{Y}, \mathbf{Z})$ has a nonzero finite orbit. The same is true on $H_1(\widetilde{Y}, \mathbf{Z})$. In particular, the genus-three, unpunctured case of the Putman–Wieland conjecture, as stated in [3, Conjecture 6.2.1], fails.*

Proof. The restriction $\beta|_N$ is nonzero by (5.2). It is fixed by $\widehat{\Gamma}$, which is a finite-index subgroup of $\mathrm{Mod}_{3,1}$ contained in the stabilizer of N. Its orbit under that stabilizer is therefore finite. The homology assertion follows by equivariant Poincaré duality on the closed oriented covering surface. □

6 Algebraic isomonodromic sections

We now relate the group-theoretic construction to families of curves. These consequences are separate from the proof of Theorem 1.1.

Intersect $\widehat{\Gamma}$ with the inverse image of a sufficiently small fine level subgroup of Mod_3, and replace Γ by its image. This does not change any invariance assertion. Let $\mathcal{T}_3$ be Teichmüller space and put

$$S = \mathcal{T}_3/\Gamma.$$

By the Riemann existence theorem, this finite analytic cover of a fine level moduli space has a unique finite étale algebraic structure. Thus S is a smooth algebraic variety, finite étale over a fine level cover of $\mathcal{M}_3$. Let $\mathcal{C}_S \to S$ be its universal curve, and write $M_{\mathrm{dR}}(\mathcal{C}_S/S, 81)$ for its relative coarse de Rham moduli space for GL_{81}.

Proposition 6.1. *For every fixed $t \in \mathbf{C}^\times$, the character $[R_t]$ determines an algebraic section of $M_{\mathrm{dR}}(\mathcal{C}_S/S, 81)$. Its analytification is horizontal for nonabelian Gauss–Manin, and its determinant class is the section of trivial connections. Infinitely many of these sections are distinct.*

Proof. The Γ-invariant character gives a single-valued horizontal analytic section by the relative Riemann–Hilbert correspondence. To prove algebraicity, we realize it after a smooth surjective base change by an algebraic connection on a total space.

Let T be the finite cover of pointed moduli corresponding to $\widehat{\Gamma}$. The map to S factors as

$$T \longrightarrow \mathcal{C}_S \longrightarrow S,$$

where the first map is finite étale and the second is smooth projective. In particular, $T \to S$ is smooth, proper, and surjective. The pulled-back curve $\mathcal{C}_T \to T$ has a tautological section. Its fundamental group is $\Pi \rtimes \widehat{\Gamma}$.

Since $\widehat{\Gamma}$ fixes the literal homomorphism ρ,

$$(\gamma, \sigma) \longmapsto \rho(\gamma)$$

defines a finite representation of $\pi_1(\mathcal{C}_T)$. The preimage of K gives a finite étale cover $\mathcal{Y} \to \mathcal{C}_T$ with

$$\pi_1(\mathcal{Y}) = H \rtimes \widehat{\Gamma}.$$

Equation (5.3) allows β to extend to this semidirect product by zero on $\widehat{\Gamma}$. For fixed t, exponentiating and taking finite direct image therefore gives a rank-81 local system on $\mathcal{C}_T$ whose restriction to each fiber has character $[R_t]$.

Algebraic Riemann–Hilbert realizes this total local system by a regular-singular algebraic integrable connection; see, for example, [1, Theorem 3.2.1]. It defines an algebraic section of the relative coarse de Rham space over T. On $T \times_S T$, the two resulting class maps agree on complex points, because $[R_t]$ is fixed by Γ. The fiber product is smooth and hence reduced, so the two algebraic maps agree. Faithfully flat descent along $T \to S$ gives the section over S.

Triviality of the relative determinant follows from $R_t(\Pi) \subset \mathrm{SL}_{81}$. Finally, the trace formula distinguishes infinitely many characters, and hence infinitely many sections. □

The parameter curve in Theorem 1.1 is algebraic in Betti coordinates. Proposition 6.1 proves algebraicity of the de Rham section for each fixed parameter; it does not assert joint algebraicity in (s, t) in de Rham coordinates.

A A two-factor calculation for the mixed twists

We prove the mixed-twist identities in Lemma 4.1 by a calculation on $U \otimes U$. This also explains why the two correction terms in (4.6) have the same origin.

A.1 Four elementary vector identities

On $U \otimes U$, use A, B for the shift and clock on the first factor, and C, L for those on the second factor. Explicitly,

$$\begin{aligned} Ae_{x,y} &= e_{x+1,y}, & Be_{x,y} &= \zeta^{-x} e_{x,y}, \\ Ce_{x,y} &= e_{x,y+1}, & Le_{x,y} &= \zeta^{-y} e_{x,y}. \end{aligned}$$

Set

$$s = u \otimes u, \qquad J = \zeta^2 L^{-1}, \qquad T = JB, \qquad Me_{x,y} = \zeta^{(x-y)(x-y+1)} e_{x,y}.$$

The ratios of successive diagonal entries give

$$MAM^{-1} = AT, \qquad MCM^{-1} = T^{-1}C, \qquad MB = BM, \qquad ML = LM. \tag{A.1}$$

Indeed,

$$\frac{M_{x+1,y}}{M_{x,y}} = \zeta^{-1-x+y}, \qquad \frac{M_{x,y+1}}{M_{x,y}} = \zeta^{x-y},$$

which are respectively the coefficients of $ATe_{x,y}$ and $T^{-1}Ce_{x,y}$.

The cochain calculation uses only the two vectors

$$f = (M - I)s = (\zeta - 1)e_1 \otimes e_0, \qquad q = (J - I)s = (\zeta^2 - 1)u \otimes e_0. \tag{A.2}$$

They satisfy

$$(AT - I)f = \zeta Aq, \tag{A.3}$$

$$\zeta T^{-1} q - f = (\zeta - \zeta^2) e_0 \otimes e_0, \tag{A.4}$$

$$(T^{-1}C - I)f = -\zeta^2 (I - B^{-1})s, \tag{A.5}$$

$$(L - I)f = 0. \tag{A.6}$$

Here every equality can be read off from at most two basis vectors. For (A.3) and (A.5), the left-hand sides are, respectively,

$$(\zeta - 1)(\zeta e_2 - e_1) \otimes e_0, \qquad (\zeta - 1)e_1 \otimes (\zeta e_1 - e_0).$$

Equation (A.4) follows by substituting

$$T^{-1}q = (\zeta^2 - 1)(\zeta e_0 - e_1) \otimes e_0,$$

and (A.6) follows from $Le_{x,0} = e_{x,0}$. In particular, applying $B - I$ to the right-hand side of (A.4) gives zero.

A.2 The uniform cocycle identity

Let z_λ be the cocycle with potentials $c_i = \lambda_i v$, where

$$\lambda_1 + \zeta\lambda_2 + \zeta^2\lambda_3 = 0.$$

Fix $k = 1$ or 2, and let j be the remaining handle. We use the preceding notation on factors $k, k+1$, tensoring every vector with u on factor j. In particular, now $f = (M_k - I)v$ and $q = (J - I)v$. Write

$$\lambda = \lambda_k, \qquad \mu = \lambda_{k+1}, \qquad \nu = \lambda_j.$$

The coefficient relation and its cyclic permutation give

$$\lambda + \zeta\mu + \zeta^2\nu = 0, \qquad \mu - \lambda = \zeta(\lambda - \nu), \qquad \mu - \nu = -\zeta^2(\lambda - \nu). \tag{A.7}$$

For $\Theta = \Theta_k$, we have $\rho(\Theta) = T$. Thus (A.1) is precisely the intertwining relation for τ_k. On the other tensor factor, the intertwiner commutes with both coefficient matrices.

The cocycle value on Θ is

$$\begin{aligned} z_\lambda(\Theta) &= (J - I)\mu v + J(B - I)\lambda v \\ &= \lambda(T - I)v + (\mu - \lambda)q. \end{aligned} \tag{A.8}$$

Define

$$w = \nu(M_k^{-1}v - v), \qquad p_1 = \lambda v + (\lambda - \nu)f, \qquad p_2 = \mu v + (\mu - \nu)f.$$

These vectors were chosen so that

$$p_1 = M_k(\lambda v + w), \qquad p_2 = M_k(\mu v + w). \tag{A.9}$$

We check the desired identity on each surface generator. For the first handle, (A.8) gives

$$\begin{aligned} z_\lambda(\tau_k(a_k)) &= (A - I)\lambda v + A z_\lambda(\Theta) \\ &= (AT - I)\lambda v + (\mu - \lambda)Aq \\ &= (AT - I)p_1, \\ z_\lambda(\tau_k(b_k)) &= T^{-1}\big((B - I)\lambda v + (B - I)z_\lambda(\Theta)\big) \\ &= (B - I)\big(\lambda v + (\mu - \lambda)T^{-1}q\big) \\ &= (B - I)p_1. \end{aligned}$$

The final equalities use (A.3), (A.4), and (A.7). For the second handle, we similarly obtain

$$\begin{aligned} z_\lambda(\tau_k(a_{k+1})) &= T^{-1}\big((C-I)\mu v - z_\lambda(\Theta)\big) \\ &= \mu(T^{-1}C - I)v + (\mu-\lambda)(I - B^{-1})v \\ &= (T^{-1}C - I)p_2, \\ z_\lambda(\tau_k(b_{k+1})) &= (L-I)\mu v = (L-I)p_2. \end{aligned}$$

Here we use $T^{-1}J = B^{-1}$, followed by (A.5), (A.6), and (A.7). By (A.1) and (A.9), these four expressions all have the form

$$M_k\big(z_\lambda(g) + (\rho(g) - I)w\big).$$

Finally, τ_k fixes a_j, b_j, whose coefficient matrices commute with M_k. The equality $M_k(\nu v + w) = \nu v$ gives

$$\begin{aligned} z_\lambda(\tau_k(a_j)) &= (A_j - I)\nu v = M_k\big(z_\lambda(a_j) + (A_j - I)w\big), \\ z_\lambda(\tau_k(b_j)) &= (B_j - I)\nu v = M_k\big(z_\lambda(b_j) + (B_j - I)w\big). \end{aligned}$$

All six generators have now been checked. After applying M_k^{-1}, both sides are cocycles for ρ, so they agree on all of Π. We have proved the uniform identity

$$M_k^{-1}(z_\lambda \circ \tau_k) = z_\lambda + \delta\big(\lambda_j(M_k^{-1}v - v)\big). \tag{A.10}$$

For $(\lambda_1, \lambda_2, \lambda_3) = (1, -\zeta^2, 0)$, the correction is zero for $k = 1$. For $k = 2$, it is

$$M_2^{-1}v - v = (\zeta - \zeta^2)u \otimes e_1 \otimes e_0,$$

as claimed in (4.6).

References

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